

A polyvinylpyrrolidone-based surface-active copolymer for an effective marine antifouling coating

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ABSTRACT

Currently, it is highly desirable to develop a stable and effective antifouling coating in maritime industries. In this work, a novel surface-active copolymer basing on hydrophilic polyvinylpyrrolidone (PVP) and hydrophobic poly (dimethylsiloxane) (PDMS) was designed and incorporated into a crosslinked PDMS matrix to form a surface-renewable antifouling coating. In the copolymer, PVP segments possessed a chemical stability and could prevent the settlement of fouling organisms attributable to its strong hydrophilicity; PDMS endowed the copolymer with compatibility to PDMS matrix and could efficiently reduce the fouling adhesion strength attributable to the low surface energy. Based on the surface-active copolymer, the coating could be reconstructed underwater in response to the environment forming a renewable surface to promote the fouling-release property. In addition, a set of controllable composition coatings were prepared to investigate the effects of surface chemistry on anti-fouling and fouling-release performance. Laboratory bioassays against marine unicellular diatom adhesion (~99 % reduction) and mimetic barnacle attachment (~81 % reduction) demonstrated an outstanding antifouling and fouling-release property of the copolymer-based coating compared to pristine PDMS, and the coating with a higher PVP content exhibited a better performance. Moreover, a long-term antifouling capability in marine field test (~4 months) was also demonstrated. This work may offer a promising solution to the problems induced by marine biofouling.

1. Introduction

Marine biofouling is an undesired accumulation of marine organisms such as microorganisms and seaweeds on submerged surfaces of marine equipment. It is a complex issue that leads to serious economic loss and unfavorable ecological impacts on marine ecosystems [1–6]. For example, marine biofouling annually costs almost 150 billion dollars in the world [7]. It can increase the hydrodynamic drag, thus decrease sailing speed and maneuverability [8–11]. Therefore, high fuel consumption is required to maintain the speed and navigation setting [12–14]. It has been reported that the increased roughness of a heavily fouled ship hull induces 45 % increase of fuel costs [15,16]. Currently, biocide-containing paints are used as the major strategy to resist

biofouling [17,18]. However, the biocide release from these paints can inevitably cause the environmental burdens [19–21].

Many non-toxic alternatives have been investigated, including surface microstructure [22–24] and self-polishing coating [25], and some antifouling coatings are incorporated the bioactive molecules [26,27] or the adhesives to replace biocide [28]. There into, poly(dimethylsiloxane) (PDMS)-based fouling-release coatings have been widely used due to their low surface energy, elastic modulus and Young's modulus [29–32]. Some marine hard foulants (such as barnacle and mussels) present a weak adhesion strength on PDMS material surfaces, resulting in the easy removal of these foulants under hydrodynamic shear forces generated by water movement [5]. However, PDMS-based coatings commonly fail to resist the diatoms (silica-based cell wall) as well as the

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slimes (containing lots of silicon) dominated by microorganism, due to the hydrophobic interaction between silica-based substance [33,34].

In recent studies, amphiphilic copolymers containing both hydrophilic (polar) and hydrophobic (nonpolar) components have been developed for antifouling and fouling-release coatings [35–38]. They incorporated hydrophilic groups (commonly used polyethylene glycol, PEG) to resist protein adhesion and hydrophobic groups (such as alkyl, fluorinated, or siloxane groups) to reduce the biofouling adhesion strength [39–42]. Amphiphilic materials exhibit an ‘ambiguous’ surface to the complex proteins or cells against their attachment, and their surfaces can be reconstructed underwater in response to environment for promoting fouling-release property [43,44]. However, it has been reported that the amphiphilic antifouling coatings based on PEG are vulnerable to oxidation, due to the breakage of ether linkage in PEG especially in the presence of oxygen and transition-metal ions, inducing the loss of antifouling property [45,46]. Therefore, more chemically stable and efficient coatings are highly desired to meet the challenge of marine biofouling.

Hydrophilic poly(N-vinylpyrrolidone) (PVP) is a biocompatible amide-containing polymer. When grafted on substrate surfaces, it presented anti-biofouling properties against protein adsorption and cell adhesion due to the formation of water barrier by polar pyrrolidone

units [38,47–49]. Moreover, PVP possesses an excellent chemical stability and is highly difficult to oxidize. For example, PVP materials can resist oxidation to maintain their native structure even in 3% of H_2O_2 solution for 100 h [50,51]. Based on the chemically stable PVP and PDMS segments, we synthesized a surface-active amphiphilic block copolymer and blended it into a crosslinking PDMS matrix to form a surface-renewable antifouling coating, as shown in Fig. 1a. In the surface-active copolymer, PVP could prevent the settlement of fouling organisms attributable to its excellent hydrophilicity, while PDMS could reduce the fouling adhesion strength and serve as a surface-directing group for the copolymer to be compatible with the PDMS matrix. Upon entering the aqueous environment, PVP segments could migrate to coating surface to bound the water molecules and resist biofouling, further inducing the reconstruction of PDMS chains and making the coating renewable. In addition, the laboratorial assays and long-term marine field test were performed to evaluate the antifouling performance of the surface-active copolymer-based coating.

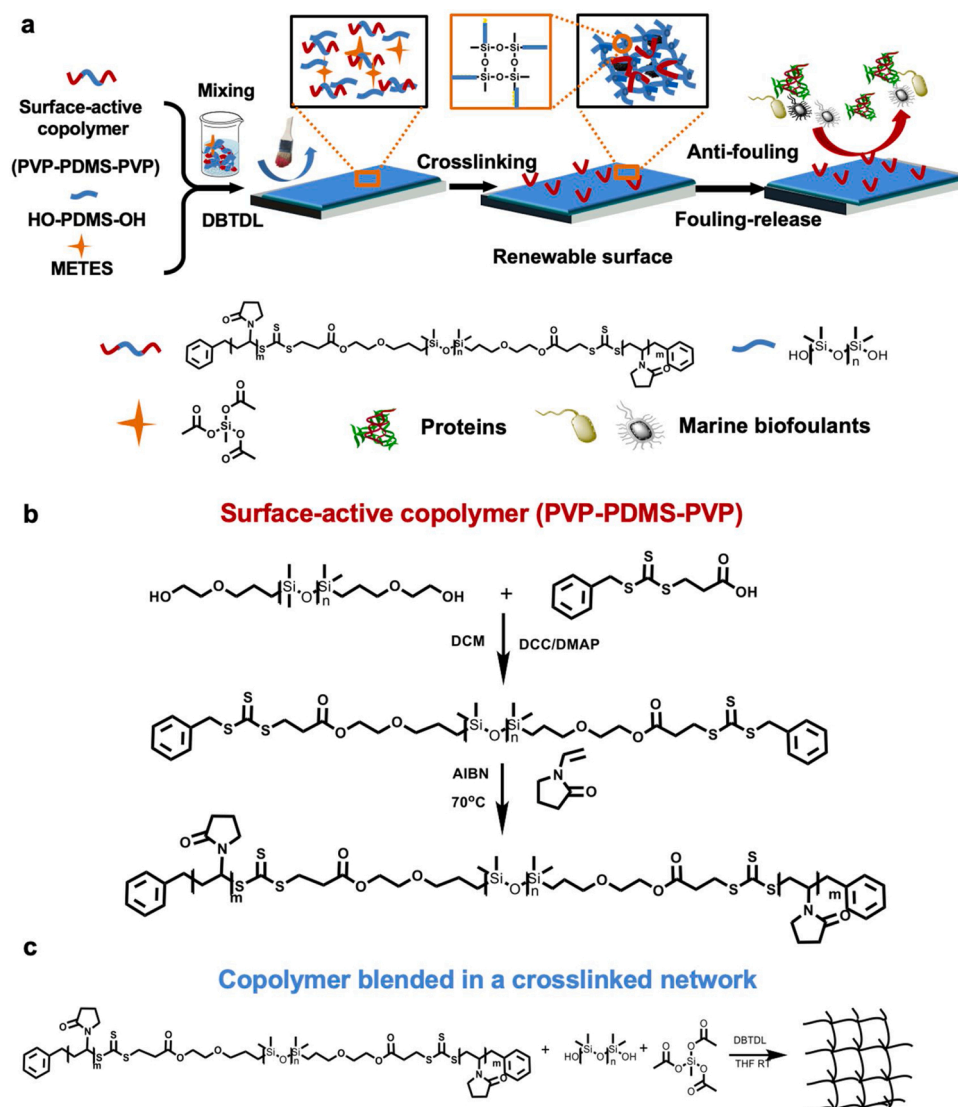


Fig. 1. (a) Schematic illustration for the design of surface-active copolymer-based antifouling coating. (b) Synthesis process of PVP-based surface-active amphiphilic copolymer and (c) the copolymer blended in a crosslinked network to form a surface-renewable coating.

2. Experimental section

2.1. Materials

Dicyclohexylcarbodiimide (DCC), tert-butyl(chloro)dimethylsilane, triethoxymethyl-silane (puried: 96 %), 4-(dimethylamino) pyridine (DMAP), dibutyltin dilaurate (DBTDL, 97.5 %), dichloromethane (DCM), hexane, methanol, and ethyl acetate were all purchased from Heowns. Two molecular weights of α,ω -dihydroxyethylpropoxyl-PDMS were obtained from Sigma-Aldrich (5.6 k HO-PDMS-OH) and Heowns (30 k HO-PDMS-OH). N-vinylpyrrolidone (NVP, Adamas, 99 % stabilized with NaOH) was purified by distillation under the reduced pressure to remove the inhibitors. Diiodomethane (CH_2I_2), 2, 2'-Azobis(isobutyronitrile) (AIBN) were both purchased from Sigma-Aldrich, and AIBN was recrystallized from methanol before use. Artificial seawater was prepared in our laboratory (ASTM standard D 1141-90). *Navicula* diatoms were obtained from Science and Technology on Marine Corrosion and Protection Laboratory, Qing dao, China. Deionized water was used in all experiments.

2.2. Synthesis of bifunctional PDMS Macro-CTA

As showed in Fig. 1b and c, the bifunctional PDMS macro-CTA was prepared by a simple DCC/DMAP esterification method according to the previous literature [52]. 3-(benzylsulfanyl thiocarbonylsulfanyl) propionic acid [53] (0.81 g, 3.0 mmol), HO-PDMS-OH (5 g, 1.2 mmol), and DMAP (0.15 g, 1.14 mmol) were dissolved in CH_2Cl_2 and stirred in an inert atmosphere. Then a CH_2Cl_2 solution contained DCC (0.6 g, 2.9 mmol) was slowly dropped into the mixture over 30 min at -10°C and allowed to react for 24 h at 30°C . Finally, the mixture was filtered and removed CH_2Cl_2 by evaporation, and the crude product was dissolved in n-hexane and washed three times each with methanol, saturated NaCl aqueous solution and distilled water. After drying with magnesium sulfate and evaporating n-hexane, the yellowish transparent liquid was obtained by drying at 40°C in vacuum overnight. ^1H NMR (CDCl_3 , δ ppm): 0.01–0.19 (m, 6H, $\text{Si}(\text{CH}_3)_2\text{-O}$); 0.53–0.59 (t, 2H, $\text{Si}(\text{CH}_3)_2\text{-CH}_2$); 1.67 (m, 3H, $\text{CH}_2\text{-CH}_2\text{-CH}_2\text{-Si}$); 2.81 (t, 2H, $\text{CH}_2\text{-CH}_2\text{-COO}$); 3.51 (t, 2H, $\text{CH}_2\text{-CH}_2\text{-CH}_2\text{-Si}$); 3.68 (m, 4H, $\text{CH}_2\text{-CH}_2\text{-COO}$ and $\text{CH}_2\text{-CH}_2\text{-O}$); 4.27 (t, 2H, $\text{CH}_2\text{-CH}_2\text{-O}$); 4.68 (t, 2H, $\text{Ar-CH}_2\text{-S}$); 7.27–7.32 (m, 5H, Ar).

2.3. Synthesis of the surface-active copolymers (PVP-PDMS-PVP)

The PVP-PDMS-PVP was prepared via RAFT copolymerization using PDMS macro-CTA. Three molar ratios of copolymers were synthesized. For P1 sample, the molar ratio of PDMS macro-CTA/NVP/AIBN was 0.625/100/1, and the molar ratios of other samples were shown in Table 1. The synthesis process of P1 as an example was presented as

Table 1
Polymerization Conditions and Physicochemical Characterization of the Surface-active Copolymers.

Copolymer	NVP/ PDMS- CTA ^a	NVP/PDMS- CTA(NMR) ^b	Mn (NMR) ^c	Mn (GPC) ^d	Mn (target) ^e
P1	138/1	135/1	20.99 kg/ mol	N/A	21.32 kg/ mol
P2	180/1	174/1	28.79 kg/ mol	N/A	29.82 kg/ mol
P3	270/1	258/1	34.64 kg/ mol	N/A	35.97 kg/ mol

^a Comonomer NVP/macoinitiator mole ratio in the initial feed.

^b Molar ratio of repeat units in the block of block copolymer by NMR.

^c Molecular weight calculated from NMR (in kg mol^{-1}).

^d GPC number-average molecular weight (in kg mol^{-1}).

^e Molecular weight calculated from target (in kg mol^{-1}).

follows: a solution containing AIBN (16.40 mg, 1.00 mmol), PDMS macro-CTA (3.7 g, 0.625 mmol), and NVP (11.11 g, 100.00 mmol) in 8 mL 1, 4-dioxane was placed into a dried flask along with a magnetic stirrer bar. Then the solution was deoxygenated by bubbling N_2 for 30 min at room temperature. The reaction flask was immersed in an oil bath preheated to 75°C for 24 h and then quenched with ice water mixture. The obtained solution was concentrated and precipitated at least three times into a tenfold cold diethyl ether. The solid was collected after filtration and then dried at 40°C in vacuum overnight. ^1H NMR (CDCl_3 , δ ppm): 0.01–0.16 (m, 6H, $\text{Si}(\text{CH}_3)_2\text{-O}$); 0.53–0.59 (t, 2H, $\text{Si}(\text{CH}_3)_2\text{-CH}_2$); 1.25–1.85 (m, 2H, $\text{N-CH}_2\text{-CH}_2\text{-CH}_2\text{-CO}$); 1.85–2.11 (m, 2H, $\text{S-CH}_2\text{-CHNC}$); 2.11–2.51 (m, 2H, $\text{N-CH}_2\text{-CH}_2\text{-CH}_2\text{-CO}$); 3.12–3.48 (m, 2H, $\text{N-CH}_2\text{-CH}_2\text{-CH}_2\text{-CO}$); 3.51–4.01 (m, 1H, $\text{S-CH}_2\text{-CHNC}$).

2.4. Preparation of the surface-active copolymer-based coatings

The coatings were prepared using a previously reported method with optimization [12]. PDMS coating was deposited in two steps onto acetone-cleaned dried glass slides ($20 \times 50 \text{ mm}^2$). In a typical preparation of film P15, the mixture composed of 5.0 g PDMS, 0.125 g cross-linker triethoxymethyl-silane and 50 mg catalyst DBTDL was dissolved in 25 mL of ethyl acetate. The resultant solution was coated on glasses, and then dried at room temperature for 24 h and annealed at 120°C for another 24 h. A solution of the same composition, containing 250 mg of the copolymer P1, was cast on top of the first layer and dried at room temperature for 48 h and subsequently at 120°C for 24 h. The surface-active block copolymers were added at concentrations of 5, 8 and 10 wt% as a fraction of the PDMS matrix. A pristine PDMS film (without copolymer) was used as a blank control.

2.5. Characterization

^1H NMR spectra were recorded on a Bruker DRX-600 instrument 400 MHz spectrometer with CDCl_3 solution; chemical shifts (δ) were reported in parts per million (ppm) relative to trimethyl silane (TMS). The ratio of PVP/PDMS was determined by comparing peaks associated with the PVP units at 3.1–3.5 ppm to the corresponding peaks of the PDMS units at 0.1 ppm to determine relative compositions. In addition, the number-average molecular weight, Mn of the copolymers was calculated from the observed peaks of the siloxane block at 0.1 and 0.5 ppm. FTIR was performed on a Nicolet FTIR 6700 spectrometer (Nexue, USA). Contact angles were measured by the sessile-drop method using an optical CA goniometry optical meter (Shanghai Zhongchen Digital Technology Apparatus Co. JC2000D1) at ambient conditions. Two types of liquids, deionized water and diiodomethane were used. The liquid droplet volumes were 5 and 2.5 μL for water and diiodomethane, respectively. The reported contact angles were an average of five individual measurements in different regions of each coating. The surface free energy of the coatings (γ_s) and its dispersive (γ_s^D) and polar (γ_s^P) components were calculated using the Owens-Wendt method [54]. The elastic modulus of silicone antifouling coatings was tested by using Shanghai Hualong Electronic Universal Testing Machine.

2.6. Attachment of *Navicula parva* diatom

Three slides were equilibrated in filtered ASW for 72 h prior to the assays. *Navicula parva* diatom suspension was prepared as described [55] with the concentration of 1×10^6 cells mL^{-1} using the filtered and sterilized seawater (the density was 1.024 g/cm^3 , pH is 8.1–8.3, and the ratio of sea salt to water was 1:30). 10 mL of cell suspension was added to the sample surfaces (three replicates of each coating) in quadriperm dishes. After 24 h, the slides were washed by seawater to remove the unattached cells, and then immediately subjected to optical microscope observation (Nikon Eclipse Ti-s). Diatoms were counted in 30 fields of view from each coating to provide the statistical average number of adhered diatoms with 95 % confidence interval.

2.7. Mimetic measurement of barnacle adhesion strength

The mimetic measurement of barnacle adhesion strength was used according to American Society for Testing and Material (ASTM) D 5618, and the PosiTest produced by DeFelsko was used to simulate the barnacle. The stud was stuck on the coatings by epoxy. After the epoxy dried, pull-test adhesion data was obtained by detaching the dolly from the substrate at speed of 4.5 N s^{-1} . Three different regions for each sample were tested. Adhesion strength in shear was calculated by dividing the measured force required to remove the stud by the basal area. Error bars represent the standard error of the mean.

2.8. Marine field tests

Marine field tests were performed from May 8, to September 16, in 2018 at the inner Qingdao bay ($120^{\circ}30'E$, $35^{\circ}35'N$) in China, where fouling organisms are diversified with a heavy biofouling pressure. The samples (thickness of the coating ranging from 250 to $300 \mu\text{m}$) applied onto carbon steel plate ($350 \text{ mm} \times 100 \text{ mm} \times 1.5 \text{ mm}$) at 25°C and 53 % of relative humidity and exposed to seawater at a depth of 0.2–2.0 m from a stationary experimental raft. The plates were taken out of sea periodically and washed carefully with seawater to observe the deposition of sea organisms and seaweed on the plate. Polyvinyl chloride (PVC) panel without any coating and coated with PDMS were used as the control. The anti-biofouling activity was evaluated according to American Society for Testing Materials (ASTM) D6990-05 (2011).

3. Results and discussion

3.1. Preparation of surface-active copolymer-based coating

As shown in Fig. 1b, difunctional macro-RAFT agent was synthesized quantitatively from the hydroxylated PDMS and CTA by an esterification method. Then the tri-block copolymer PVP-PDMS-PVP were synthesized via RAFT polymerization using NVP and CTA-PDMS-CTA as the macro-chain transfer agent. Here, we synthesized three molar ratios of PVP based copolymer as presented in Table 1. Copolymers were identified by a shorthand name P1, P2, or P3.

^1H NMR was used to confirm the chemical structure of 3-(benzylsulfanyl thiocarbonylsulfanyl) propionic acid, CTA-PDMS-CTA and copolymers, as shown in Fig. 2. Fig. 2a showed the spectrum of 3-(benzylsulfanyl thiocarbonylsulfanyl) propionic acid as previously described. In the spectrum of CTA-PDMS-CTA (Fig. 2b), the resonance at 3.73 ppm associated with $-\text{CH}_2-\text{OH}$ in OH-PDMS-OH was disappeared, and two new signals at 3.6 ppm and 4.3 ppm were attributed to the $-\text{CH}_2-\text{S}-$ and $-\text{CH}_2-\text{O}-$ group (g and f), respectively. Moreover, the integral ratio of two peaks was approximately 1:1, indicating that $-\text{OH}$ end groups essentially converted into the macro-chain transfer agent CTA-PDMS-CTA. In Fig. 2c, the signals at 0.1 ppm were contributed by the methyl protons of PDMS. The δ 1.3–1.8 was the proton absorption peak of the methylene group at the 4-position of the pyrrolidone ring $-\text{CH}_2-$ (l). δ 1.8–2.1 was the proton absorption peak of methylene on the main chain $-\text{CH}_2-$ (n_1). δ 2.1–2.5 was the absorption peak of the methylene proton adjacent to the pyrrolidone ring $-\text{CO}-$ bond far away from the pyrrolidone ring (k). δ 3.0–3.4 was the absorption peak of the methylene proton adjacent to the N bond on the pyrrolidone ring $-\text{NCH}_2-$ near the pyrrolidone ring (m_1); δ 3.7–4.0 was the proton peak of the methine connected to N on the pyrrole backbone (o). The molecular weight was calculated by the peak at 3.0–3.4 ppm attributable to PVP and peak at 0.1 ppm attributable to PDMS. These results indicated the successful synthesis of chain transfer agents CTA-PDMS-CTA and copolymer PVP-PDMS-PVP.

In the FTIR spectra, the characteristic peaks of the copolymer were all presented in Fig. 3. The absorption peak in the range of $1000\text{--}1150 \text{ cm}^{-1}$ was assigned to Si-O-C asymmetric stretching vibrations that appeared in all the samples. In addition, the peak detected at 795 cm^{-1} was attributed to the Si-O-Si bending vibration, which was a characteristic signal of PDMS. Because PVP possesses a strong hydrophilicity, the polymer contains a small amount of bound water, inducing water absorption peak at 3418 cm^{-1} . The peaks in 1659 cm^{-1} , 1270 cm^{-1} , and 2962 cm^{-1} were stretching-vibration peak of carboxyl group $-\text{CO}-$, $-\text{CN}-$, and $-\text{CH}-$, respectively [56]. The results of FTIR spectra demonstrated the successful polymerization of copolymer PVP-PDMS-PVP.

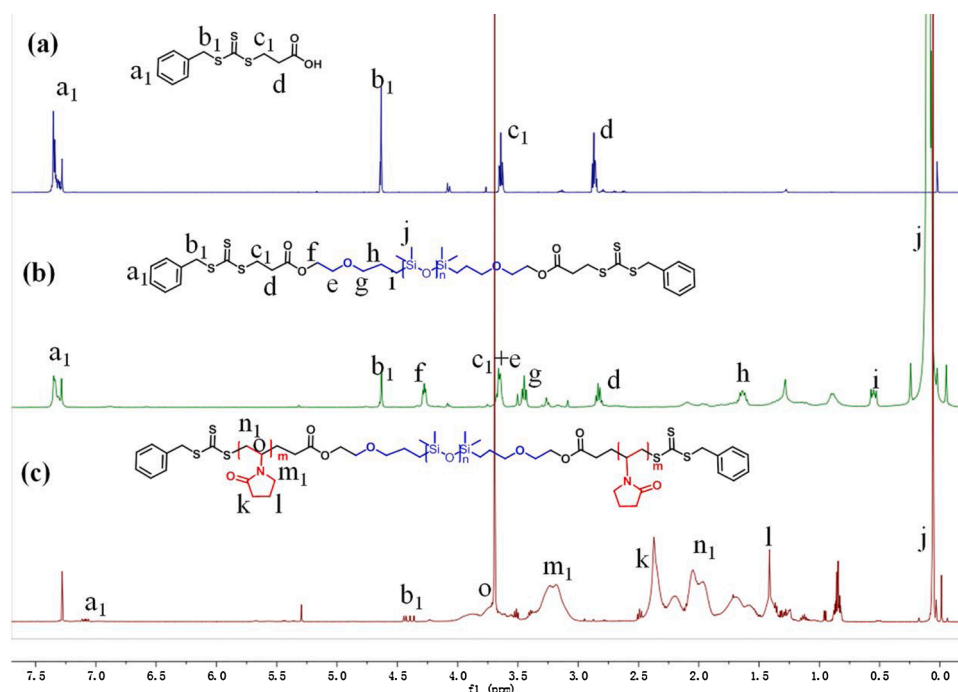


Fig. 2. ^1H NMR spectra of (a) RAFT agent 3-(benzylsulfanyl thiocarbonylsulfanyl) propionic acid (b) PDMS macro-CTA (c) PVP-PDMS-PVP.

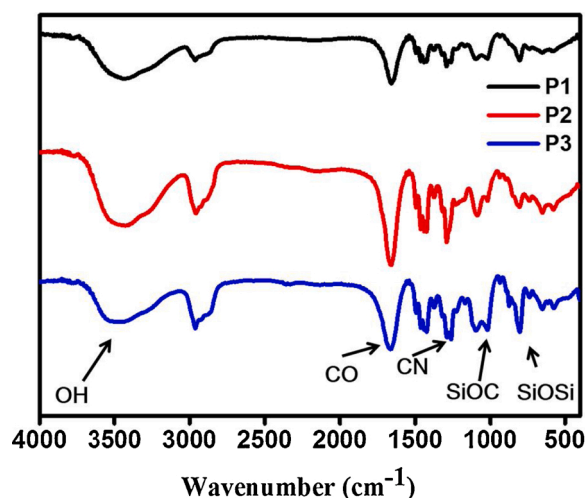


Fig. 3. Fourier transform infrared spectra (FTIR) of copolymer P1, P2 and P3.

3.2. Static contact angle, surface free energy, and elastic modulus

To systematically investigate the effects of the copolymer on anti-fouling and fouling-release properties, we developed the different molecular weights, ratios, and contents of copolymers to test the static water, diiodomethane contact angles, surface free energy, and elastic modulus of coatings, as shown in Table 2, Fig. 4, and Tab.S1.

Firstly, water contact angle measurements were carried out immediately after immersing the coatings in water. As shown in Table 2 ($t = 0$), the contact angle of surface-active copolymer-based coating surfaces (97.1° – 110.2°) were all lower than that of PDMS (112.6°), suggesting the more hydrophilic coating surfaces induced by the addition of copolymers. In addition, the contact angle of these coatings presented an inverse proportion to the content of copolymer. Moreover, surface free energy of coatings were calculated by two different probe liquids (H_2O and CH_2I_2) with known surface tensions, according to Owens-Wendt method [57–59]. It could be observed a higher surface free energy (γ_{SV}) of copolymer-based coatings compared with the pristine PDMS coating, attributable to the increased surface hydrophilicity induced by PVP segments. Notably, their surface free energy maintained a low

Table 2
Static Contact Angles and Surface Free Energies of Copolymer-based Coating.

Coating	Copolymer (wt%)	static contact angle (deg)		surface free energy ($mJ\ m^{-2}$)		
		θ_{H_2O}	$\theta_{CH_2I_2}$	γ_{SV}	γ_{SV}^d	γ_{SV}^p
PDMS	0	112.6 ± 1.5	70.3 $\pm$ 2.1	23.3	23.3	0.0
Polymer 1 (P1)	P15	109.8 ± 1.5	65.6 $\pm$ 1.2	26.0	26.0	0.0
	P18	106.8 ± 1.5	63.3 $\pm$ 1.2	27.1	27.0	0.1
	P110	100.6 ± 1	61.2 $\pm$ 1.3	27.8	27.3	0.5
	P25	110.2 ± 0.6	66.6 $\pm$ 0.8	25.3	25.3	0.0
Polymer 2 (P2)	P28	107.6 ± 1.6	64.2 $\pm$ 1.5	26.7	26.6	0.1
	P210	100.9 ± 1.4	62.3 $\pm$ 0.2	27.3	26.7	0.6
	P35	106.7 ± 1.4	63.8 $\pm$ 0.7	26.8	26.7	0.1
	P38	100.6 ± 2.9	61.3 $\pm$ 2.3	27.9	27.3	0.6
Polymer 3 (P3)	P310	97.1 $\pm$ 1.1	59.2 $\pm$ 0.9	29.0	28.0	1.0

surface energy of $\sim 27\ mN\cdot m^{-1}$ with no obvious increase when the total weight percentage of copolymer higher than 8%. Previous literatures have reported that barnacles were difficult to attach on the surfaces with the surface energy lower than $30\ mN\cdot m^{-1}$ [60], indicating the fouling-release performance of these copolymer-based coatings.

To study the time-dependent surface wettability, these coatings were exposed to water after the initial immersion at different times ($t = 0, 12, 24, 48, 72, 96, 120, 144$ and $168\ h$). As shown in Fig. 4, the contact angle of PDMS coating was almost constant with increasing time, while those of copolymer-based coatings decreased after a continued immersion in seawater, and the most reached an equilibrium after 24–48 h immersion. It indicated the surface reconstruction and renewal due to the migration of PVP segments to the coating surface. In Fig. 4a–c, 10 % of copolymer-based coating showed a greatest decrease on contact angles. To a similar copolymer ratio in Fig. 4d, the contact angles of the high-molecular-weight copolymer-based coatings (P2 and P3) were much lower than those of the coating based on the low molecular weight copolymer (P1) after 168 h immersion. Because a higher molecular weight copolymer contained a relative high percentage of hydrophilic PVP. These results indicated that the coating surface could be reconstructed due to the surface-active copolymers and the reconstructive effectiveness was related to the content of PVP.

Apart from the surface energy, elastic modulus related to fouling-release property is also an important factor of silicone-based coatings. The elastic modulus of coating is proportional to the energy that is required by adherent marine fouling removal. As shown in Tab. S1, we used an electronic universal testing machine to test the elastic modulus (1.7–3.3 MPa, average: 2.5 MPa) of surface-active copolymer-based coatings, comparable to the 2.4 MPa of PDMS coating, indicating the maintained elastic properties of coatings for fouling-release performance.

3.3. Diatom settlement

Diatom, a microscopic unicellular marine colonial alga impregnated with silica, can be easy to attach on PDMS material surface via hydrophobic interaction between silica and ‘Si-C’ or ‘Si-O’. In this work, we used diatoms (*Navicula parva*) to test antifouling and fouling-release performance of copolymer-based coating surfaces, as presented in Fig. 5, Fig.S1 and S2. All copolymer-based coating surfaces had the significantly lower amount of adhered diatoms than the pristine PDMS coating surface in Fig. 5a and Fig.S1, in accordance with the previous reports that diatoms could attach on hydrophobic rather than hydrophilic surfaces [61,62]. They suggested the excellent antifouling property of surface-active copolymer-based coating, and highlighted the importance of PVP segments in the coatings [63,64]. After being subject to a low shear stress ($\sim 8\ Pa$) for 5 min, negligible cell attachment on copolymer-based coating surfaces especially on P110, P210, and P310, which presented a 99 % reduction attachment compared to PDMS, indicating the best fouling-release performance of coating at the highest PVP content (Fig. 5 and Fig.S2). These results suggested that the surface-active copolymers could improve both antifouling and fouling-release performance of the coating, attributable to a water barrier formation by migration of hydrophilic PVP to coating surface.

3.4. Pseudobarnacle adhesion

Mimetic barnacle adhesion strength measurement was accessed according to American Society for Testing and Material (ASTM) D 5618. Fig. 6 showed that the significantly lower adhesion strength of copolymer-based coating compared with that of pristine PDMS coating. Moreover, the adhesion force of the simulation fouling substance on coating surfaces exhibited a decrease with the increased contents of surface-active copolymers. Using P1 as an example, when the content of copolymer was increased to 10 %, the adhesion force on coating exhibited an 81 % reduction compared with the pristine PDMS coating

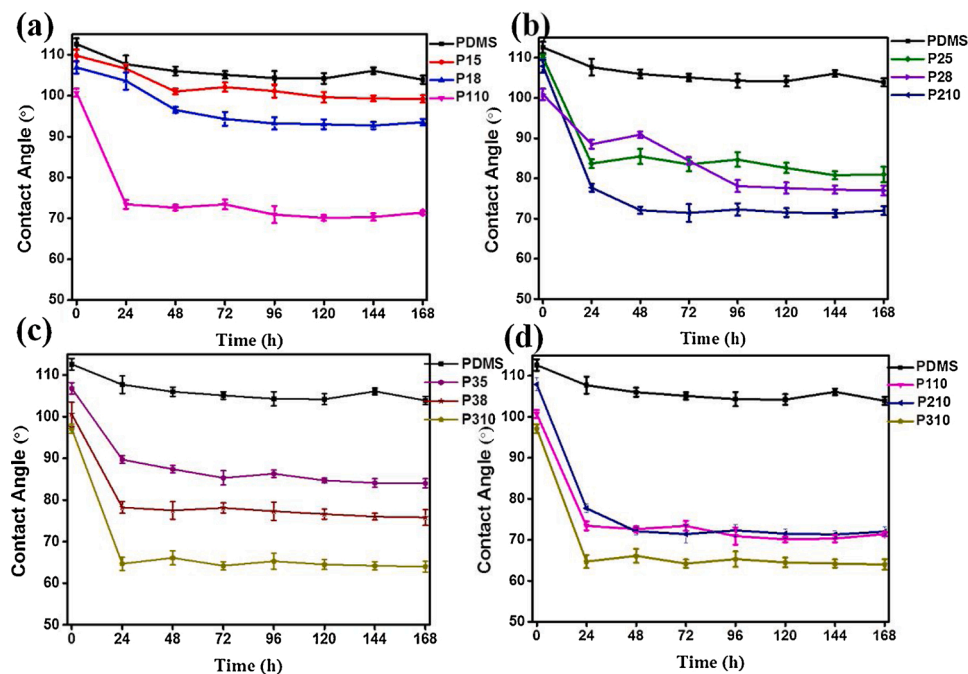


Fig. 4. Contact angle measurements of coating surfaces taken under water over 168 h immersion in water. (a) P1 (b) P2 (c) P3 and (d) P110-P310 based coatings.

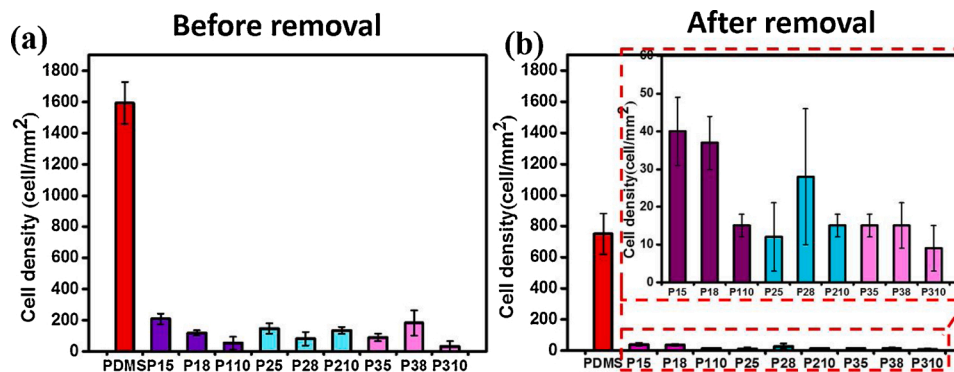


Fig. 5. (A) *Navicula parva* adhesion on pristine PDMS and surface-active copolymer-based coatings before exposure to flow. (B) *Navicula parva* adhesion on coatings after 5 min exposure to flow with a shear stress of ~ 8 Pa. Each sample is tested in 6 replicates.

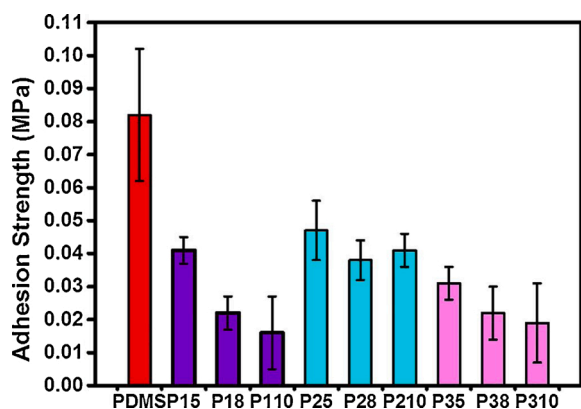


Fig. 6. The mimetic barnacle adhesion strength on coatings.

(100 %). Moreover, combined with diatom attachment results, it could be concluded that P3 copolymer-based coating had a best antifouling and fouling-release performance in laboratory assays, also consistent

with the results of contact angle test. These results indicated an easy removal of attached marine fouling on copolymer-based coatings depended on a low ship speed or a little mechanical force.

3.5. Long-term marine field test

Because of the best performance of P3-based coating in laboratory antifouling tests, P35 P38 P310 coating panels compared with pristine PDMS coating and control PVC panels were examined in a long-term marine field test (123 days). Fig. 7 showed the typical images of the tested panels before and after immersion in natural seawater in Qingdao bay (120°30'E, 35°35'N) in China with a heavy fouling pressure. After 123 days, the blank control PVC panel and pristine PDMS coating surfaces were fully covered by fouling organisms including barnacles, mussels and sponges, while P35 coated surface was covered by a significantly reduced soft fouling organisms or silt. Surprisingly, almost no contaminant on P38 and P310 were presented, demonstrating the excellent antifouling performance in even long-term marine field test.

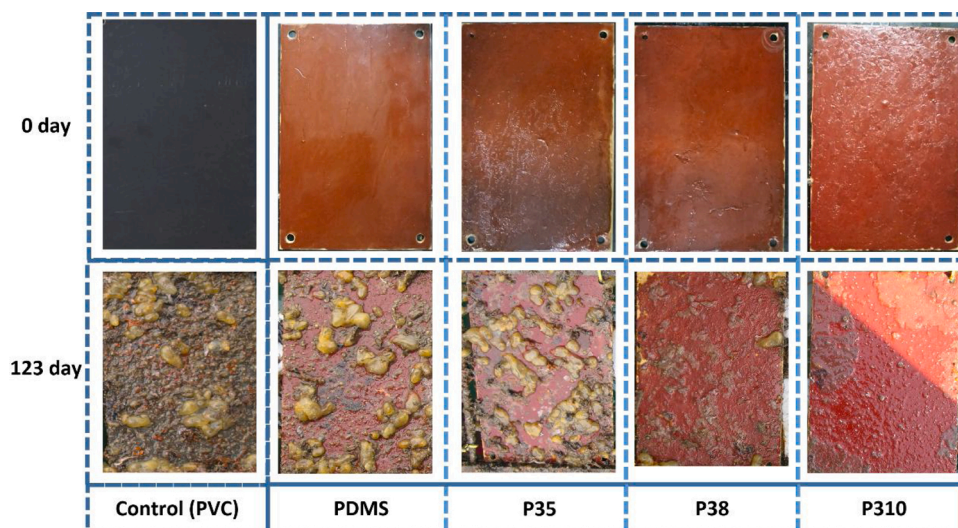


Fig. 7. Images of the tested panels before and after immersion in seawater for 123 days.

4. Conclusion

In this work, we developed a novel marine antifouling coating based on a surface-active PVP-PDMS-PVP triblock copolymer, and systematically studied the effects of molar weight, ratio, and content of copolymers on the antifouling properties. The copolymer blended in the PDMS matrix could form a renewable coating surface with the hydrophilic PVP moieties to prevent fouling attachment. Laboratorial assays presented a superior antifouling and fouling-release property compared to the pristine PDMS coating, and demonstrated the coating with a higher PVP content showed a higher efficacy on inhibiting marine organism attachment. More importantly, a long-term marine field test was performed to verify the excellent antifouling performance of this surface-active copolymer-based coating in a practical marine environment. This work could offer a promising approach to develop an environmentally-friendly, chemically stable, and effective antifouling coating, which would benefit the marine industry.

Author contributions

H. S. G., X. M. L., J. Y. and L. Z. conceived the study. All the authors performed experiments. The manuscript was written by H. S. G., X. M. L., J. Y. and L. Z. and proofread by all authors.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

CRediT authorship contribution statement

Hongshuang Guo: Conceptualization, Methodology, Writing - original draft, Investigation. **Xinmeng Liu:** Methodology, Writing - original draft, Investigation, Data curation. **Weiqliang Zhao:** Data curation. **Changhai Xie:** Data curation. **Yingnan Zhu:** Data curation. **Chiyu Wen:** Data curation. **Qingsi Li:** Data curation. **Xiaojie Sui:** Data curation. **Jing Yang:** Conceptualization, Methodology, Writing - original draft, Investigation, Writing - review & editing. **Lei Zhang:** Conceptualization, Methodology, Writing - original draft, Supervision, Writing - review & editing.

Declaration of Competing Interest

The authors declare no competing financial interest.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.porgcoat.2020.105975>.

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